

Which Weather Indicators Matter Most for Public Communication in Malaysia?

ADS1001 MWF-G2 | Client: Malaysian public weather information service | Group MWF-G2: [add member names]

Big takeaway

Use indicators the public can act on immediately.

1Member 1: Client Problem

Client: a Malaysian public weather information service for residents, commuters, outdoor users and event organisers.

Problem: many raw weather variables are available, but not all are easy for the public to understand or act on.

Research question: Which indicators best communicate comfort, rainfall, wind risk and UV exposure?

2Member 2: Data Pipeline

ADS1001 Malaysia weather records

Selected Kuala Lumpur + Pulau Pinang locations

2018-01-17 to 2023-08-30

RAW57,475 rows19 columns

clean

CLEAN51,693 rows38 columns

0duplicate rows

0missing in main columns

7engineered features

COREDecision Matrix: Public Communication Value

Lead with simple, actionable measures

The best public indicators are understandable, actionable, useful and supported by data.

MAIN PUBLIC-FACING INDICATORS

Temperature

Humidity

Precipitation

Wind speed

Gust

UV categories

SUPPORTING / INTERNAL

Wind chill

Dew point

Pressure

Final answer: prioritise temperature, humidity, precipitation rate/total, wind speed, gust and UV index categories.

3Member 3: Outdoor Comfort Evidence

Method: correlation + difference analysis

0.999temperature vs wind chill correlation

0.002mean temp difference, deg C

0.284humidity vs dew point correlation

Temperature and wind chill nearly overlap

Humidity is more intuitive for public comfort

Conclusion: wind chill adds little routine value here (median difference 0 deg C). Lead public comfort messages with temperature and humidity; use dew point as technical support.

4Member 4: Risk & Warning Evidence

Method: grouped rates + warning categories

25.9%rain event rate

0.046pressure vs precipitation corr.

0.600wind speed vs gust corr.

Rain event rate by pressure quartile

Gust captures sudden wind risk

UV categories translate risk

Conclusion: precipitation is clearer than pressure for rain alerts. Communicate gust separately when sudden wind risk matters. UV is strong because categories map directly to public action (high UV >= 6: 8.8%).

Courts: 41.0k low | 6.1k moderate | 2.5k high | 1.6k very high | 0.5k extreme

5Member 5: Recommendation & Next Steps

Recommended public message stack

ComfortTemperature + humidityWind chill / dew point as context

RainPrecipitation rate or totalPressure for internal context

WindWind speed + gustGust warning for sudden risk

UVStandard UV categoriesTime-of-day support

Limitations

Selected KL and Pulau Pinang locations only; missing values and imputation may affect patterns; uneven location representation; associations are not causation; public understanding was not directly surveyed.

Future work

expand locations; test messages with public users; add forecasting models.

Presentation Handoff: same story, five equal 2-minute turns

M1 0:00-2:00

Motivation, client problem, research question.

M2 2:00-4:00

Dataset, cleaning decisions, engineered features.

M3 4:00-6:00

Comfort evidence: temperature, humidity, dew point.

M4 6:00-8:00

Warnings: precipitation, gust, UV categories.

M5 8:00-10:00

Recommendation, limitations, future work, Q&A bridge.

References: ADS1001 Malaysia weather dataset; ADS1001 Project Guidelines and Rubric; standard UV index public health categories.

A1 landscape poster | Print at full size